

Project Documentation

Objective

The objective of this project was to learn how Python can communicate with external mail servers and automate email transmission.

Concepts Learned

SMTP Protocol

SMTP (Simple Mail Transfer Protocol) is the standard protocol used to send emails over the Internet.

The script uses:

```
smtplib.SMTP()
```

to establish communication with an SMTP server.

TLS Encryption

TLS (Transport Layer Security) encrypts communication between the client and the mail server.

Implementation:

```
server.starttls()
```

Without TLS, credentials could be transmitted insecurely.

Authentication

The application authenticates the sender before sending messages.

Implementation:

```
server.login(email, password)
```

This demonstrated how applications identify themselves to external services.

File Handling

The project reads information from external files:

- password.txt
- message.txt

Implementation:

```
with open(filename, "r") as f:
```

This reinforced proper file management using context managers.

MIME Messages

Emails containing attachments must be formatted as MIME messages.

Implementation:

```
msg = MIMEMultipart()
```

MIME allows combining:

- Text content
- Attachments
- Different content types

inside a single email.

Binary File Processing

Attachments are opened in binary mode:

```
with open(file, "rb")
```

This allows non-text files such as images and PDFs to be transmitted correctly.

Base64 Encoding

Attachments are encoded before transmission:

```
encoders.encode_base64()
```

Encoding converts binary data into a format suitable for email transmission.

Program Structure

The application was organized into logical sections:

1. Configuration
2. Authentication
3. Message Creation
4. Attachment Processing
5. Email Transmission

This improved readability and maintainability.

Challenges Encountered

- Incorrect SMTP port selection.
 - MIME import mistakes.
 - Authentication errors.
 - Handling attachments correctly.
 - Understanding email encoding.
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Skills Acquired

- Python networking fundamentals.
 - SMTP communication.
 - Secure authentication.
 - File management.
 - MIME formatting.
 - Email automation.
 - Code documentation and organization.
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Conclusion

This project provided practical experience with Python's networking and email libraries. It demonstrated how applications communicate with remote services, securely authenticate users, process files, and automate repetitive tasks through scripting.